

Computer vision

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1.
Ans

Computer vision enables algorithm and AI to always boost the system and identify objects patterns and scenes.

a) objectives

- * image understanding
- * object detection and recognition.
- * Decision making

b) Applications

- * Face recognition.
- * Self driving car

2.
Ans

Image acquisition; preprocessing, feature extraction information. and image classification

a) Feature extraction: It is the process of identifying important information (edges, corners, textures) from an image.

b) object recognition it identifies and classifies objects. making computer vision subjects clear

3.
Ans

a) Low-level Image processing Basic operations like noise removal, filtering and image enhancement

b) Midlevel vs High level;

* Mid level : Segmentation & feature extraction

* High level : object recognition and scene understanding

4.
Ans

A digital image is represented as a grid of pixels

(a) pixel : the smallest unit of a digital image

(b) Image resolution : Higher resolution means more pixels and better image quality

5.
Ans

A camera sensor : It converts light and converts it into a digital image

a) Role of Image sensor :

It converts light into electrical / digital signal

b) image acquisition devices

- * Digital camera
- * Scanner

6.
Ans

a) sampling: Converts a continuous image into a discrete pixels.

b) The purpose of quantization in image processing assigns intensity values to each pixel.

*
Ans

Ans

It explains how light from a scene passes through the camera lenses and it is the process of capturing images.

a) segmentation & feature extraction that helps took place is called perspective projection

b) image processing

* It preprocesses the image, feature extraction information and image classification.
Computer vision.

* computer vision is the process of identifying important information.

8. Ans Image acquisition, preprocessing, feature extraction.

Information and image classification

a) Feature extraction: It is the process of identifying important information (edges, corners, textures) from an image.

b) object recognition It identifies and classifies objects

9. Ans A digital image represented as grid of pixels

(a) Application in healthcare.

The smallest unit of a digital image

(b) Application in autonomous vehicles

Higher resolution means more pixels and better image quality

10. Ans

A camera sensor: It converts light into digital image.

a) Role of image sensor: Converts light into electrical / digital signals

b) image acquisition devices.